
Topic: Tuples, Lists, Aliasing, Mutability, Cloning Date: 2026-09-13

① Check My Understanding

Questions completed: 5 / 5

Answers revised after AI hints: 4 / 5

② My Misconception

Before: I thought...

Cloning a list meant that everything inside the list would also be recursively duplicated, so modifying the original list would never affect the cloned list.

Now: I understand...

`[ : ]` creates a new outer list, but it is only a shallow clone. If the list contains mutable objects such as nested lists, those objects can still be shared between the original and the cloned list. A deep clone is different because it recursively duplicates the nested objects.

I also understand the distinction between mutation and reassignment. Mutation changes the existing object, while reassignment makes a variable point to a different object.

③ Challenge the AI

One AI-generated question I challenged:

“If you clone a list, then modifying an element of the original list can never affect the cloned list.”

Why?

- ☒ Ambiguous
- ☐ Oversimplified
- ☐ Technically questionable
- ☐ Too easy
- ☐ Other: _____

Brief explanation:

The question did not specify whether “clone” meant a shallow clone or a deep clone. But i think that is a great reservation. I initially assumed that cloning meant recursively duplicating everything and fell into the trap.

④ One-Minute Reflection

One thing I am still unsure about:

How a deep clone being done in python, is there any abbreviated code similar to `[ : ]`?